DisPeD: A Controlled Study of Encoder Adaptation for Multi-Turn Jailbreak Detection

Anonymous ACL submission

Abstract

Multi-turn jailbreak attacks spread adversarial intent across benign-seeming turns. Existing defenses fail to generalize: Llama Guard 3 detects only 1.8% of multi-turn jailbreaks in our evaluation. Detecting such attacks benefits substantially from domain adaptation at base model scale, but the form of adaptation matters. We isolate the factors behind cross-attack generalization through controlled ablation: thirteen encoder configurations at base scale (125M–184M parameters) on synthetic multi-turn data decompose contributions of domain exposure, classification supervision, and label granularity across three out-of-distribution (OOD) attack families. On synthetic data, classification-based adaptation recovers OOD detection (F1 = 0.97–1.0 vs. vanilla 0.26–0.31; attack-side OOD, IID benign upper bound) while unsupervised pretraining collapses; under mixed-LLM training, template attacks remain detectable but strategy-diverse attacks degrade to chance level (F1 = 0.504). On human-authored single-turn data (ToxicChat), this hierarchy reverses: masked language modeling outperforms 9-class classification (F1 0.837 vs. 0.596), evidence that classification adaptation's advantage is condition-specific rather than universal. Orthogonal to attack-side generalization, few-shot benign adaptation reduces false-positive rates from 58–68% to 0–1.5%, with 24.3× inference speedup over LLM-as-judge baselines.

1 Introduction

Multi-turn jailbreak attacks distribute adversarial intent across individually benign turns. Crescendo (Russinovich et al., 2025), Foot-in-the-Door (Wang et al., 2024), and Deceptive Delight (Palo Alto Networks Unit 42, 2024) all rely on persuasion as their core mechanism: Zeng et al. (2024b) catalogue 40 persuasion techniques across jailbreak families, and Hagendorff et al. (2026) identify strategies in 97% of successful attacks. Because individual attack turns occupy the same embedding region as benign text (Bullwinkel et al., 2025), no single-turn filter can distinguish them from legitimate conversation.

Existing defenses leave a critical gap. Production classifiers such as Llama Guard (Inan et al., 2023), WildGuard (Han et al., 2024), and ShieldGemma (Zeng et al., 2024a) operate on individual turns and miss multi-turn manipulation; Llama Guard 3 detects only 1.8% of multi-turn jailbreaks in our evaluation (Section 4.3). Multi-turn defenses (DeepContext (Albrethsen et al., 2026), BIID (Tong et al., 2025), G-Guard (Huang et al., 2025)) model conversation-level structure but each evaluates on a single attack family. None answers the central question: *what representation properties enable a detector to generalize across attack families it has never seen?*

Pretrained language understanding does not suffice at base model scale: vanilla DeBERTa-v3 (He et al., 2021) with temporal aggregation (cf. DeepContext) drops from F1 = 1.000 in-distribution to 0.312 on OOD Deceptive Delight, and GPT-5.5 reaches only 0.191–0.449 F1 (across prefix lengths) with 30–47% unparseable responses (Appendix A). The bottleneck is distributional: cross-attack generalization at base model scale benefits substantially from domain adaptation.

Our pipeline, **DisPeD** (Disentangled Persuasion Detection)[1], separates encoder adaptation (Stage 1) from temporal classification (Stage 2, a frozen Bi-GRU; Figure 1). Seven Stage 1 variants each isolate a single factor while holding all others constant (Section 3); the contribution is the decomposition, not the pipeline.

Controlled ablation across DeBERTa-v3 and RoBERTa (Liu et al., 2019b) on four attack families

[1]We use 'disentangled' to describe our experimental methodology of isolating individual factors through controlled ablation, distinct from disentangled representation learning (β-VAE, InfoGAN).

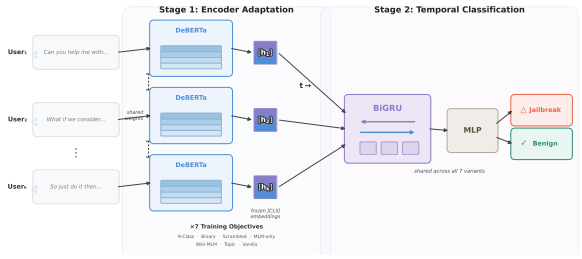


Figure 1: DisPeD architecture. Stage 1 adapts a DeBERTa encoder on jailbreak conversation text (seven variant objectives). Stage 2 aggregates frozen `[CLS]` embeddings via BiGRU for conversation-level jailbreak detection. Only user turns are processed.

and two training compositions shows that for base-size encoders, classification-based adaptation is the most robust variant on synthetic multi-turn OOD attacks. MLM collapses on mixed-LLM training (F1 = 0.289 ≈ vanilla) while classification DAPT maintains 0.818; RoBERTa-large (355M) handles ActorAttack (Ren et al., 2025) without adaptation. Classification supervision is associated with substantially lower prediction variance on synthetic data but not on human-authored conversations (Section 4.6).

Our contributions:

- **Systematic ablation of cross-attack generalization factors.** On synthetic multi-turn data, thirteen encoder configurations at three scales (125–355M) decompose jailbreak detection into two orthogonal axes (attack-side and benign-side), the first, to our knowledge, such controlled factorial study in this domain. On the attack side, classification DAPT recovers OOD detection (vanilla 0.26–0.31 → 0.97–1.0) while unsupervised pretraining collapses under mixed-LLM training. On the benign side, 10–25 target-domain conversations reduce FPR from 58–68% to 0–1.5% on two independent sources (WildChat, OASST2; IID evaluation) without detection degradation.

- **Partition sufficiency with an observed difficulty gradient (at base model scale).** Domain-relevant classification objectives, including scrambled labels, suffice for cross-attack generalization on template attacks (TOST-confirmed, $|d| \leq 0.14$ on DD, $d \approx 0$ on FITD at $n = 10$), while attacks employing more varied conversational strategies (ActorAttack) show a moderate label effect ($d = 0.44$). This is consistent with Zhang et al. (2021)'s framework: domain-relevant partitioning appears necessary (topic labels fail) and compression localizes to 3 SVD dimensions in layers 8–12. A probe over these dimensions confirms sufficiency; removing them degrades DD by 17 pp while FITD is unaffected (partial necessity; Appendix S.1).

- **Two-step adaptation protocol.** On the synthetic benchmark, continued pretraining achieves 24.3× inference speedup over LLM-as-judge (a deployment cost comparison; parameter counts differ by 43×; Appendix V); 10–25 target-domain benign conversations reduce FPR below deployment thresholds on both WildChat and OASST2 crowdsourced conversations. No target-domain attack data is required.

2 Related Work

Multi-turn jailbreak attacks. All multi-turn jailbreak families share a persuasion substrate (Zeng et al., 2024b; Hagendorff et al., 2026; Li et al., 2024), yet each operationalizes it differently: Crescendo (Russinovich et al., 2025) gradually escalates, FITD (Wang et al., 2024) exploits commitment-consistency bias, DD (Palo Alto Networks Unit 42, 2024) combines emotional manipulation with role assignment, ActorAttack (Ren et al., 2025) steers through multiple personas, and automated methods (PAIR (Chao et al., 2025), TAP (Mehrotra et al., 2024)) iteratively refine adversarial prompts, extending the automated red-teaming paradigm (Perez et al., 2022). Wei et al. (2023) provide a unifying taxonomy; RED QUEEN (Jiang et al., 2024) constructs concealed multi-turn scenarios across 14 harmful categories; benchmarks (Mazeika et al., 2024; Chao et al., 2024) standardize single-turn comparison, while MultiBreak (Song et al., 2026) targets multi-turn evaluation. This shared substrate with diverse surface strategies defines our evaluation: a detector must generalize across families, not merely succeed on its training distribution.

Multi-turn defenses and production safety. No existing defense evaluates cross-attack transfer. Recent conversation-level systems pair fine-tuned BERT (Devlin et al., 2019) encoders with sequence-level aggregation via GRU hierarchies (DeepContext (Albrethsen et al., 2026)), turn-intention inference (BIID (Tong et al., 2025)), graph neural networks (G-Guard (Huang et al., 2025)), or predictive generation (SafeDream (Yan et al., 2026)), yet each evaluates on a single attack family. Di-

rect experimental comparison with DeepContext, BIID, and G-Guard is precluded by the absence of public implementations and incompatible evaluation protocols; Appendix Q provides a parameter-matched BERT-base proxy as a partial substitute. Representation-level approaches include X-Boundary (Lu et al., 2025), JBShield (Zhang et al., 2025) via activated concept analysis, gradient-space analysis (GradSafe (Xie et al., 2024)), circuit breakers via representation engineering (Zou et al., 2024), latent activation probing (Kulkarni, 2026; Kadali and Papalexakis, 2025, 2026), and vision-language detection (Hua et al., 2026); these require white-box access and can be brittle under adaptive attacks (Derya and Sunar, 2026). Our work uses black-box classification with systematic cross-attack OOD evaluation. Per-turn classifiers (Llama Guard (Inan et al., 2023; Meta AI, 2024), ShieldGemma (Zeng et al., 2024a), WildGuard (Han et al., 2024), Constitutional Classifiers (Sharma et al., 2025)) and production content moderation systems (Markov et al., 2023) cannot model multi-turn trajectories, and perturbation-based defenses (Robey et al., 2024) and safety-aware decoding (Xu et al., 2024) target single-turn token-level attacks. Our ablation fills this gap: decomposition across thirteen configurations under OOD evaluation identifies domain exposure as the dominant factor behind cross-attack transfer.

Computational persuasion and discourse. Persuasion and discourse frameworks offer interpretable vocabulary for attack strategies, but whether such structure aids detection has not been empirically disentangled. Foundational work (RST (Mann and Thompson, 1988), PDTB (Prasad et al., 2008), dialogue acts (Stolcke et al., 2000), Speech Act Theory (Austin, 1962; Searle, 1969)) models pragmatic structure at different granularities. Building on Cialdini's principles (Cialdini, 2001), PAP (Zeng et al., 2024b) identifies 40 persuasion techniques across jailbreak families, and Hagendorff et al. (2026) find identifiable strategies in 97% of successful attacks; propaganda analysis (Da San Martino et al., 2019, 2020) and persuasive dialogue (Wang et al., 2019) apply similar technique-level classification. Our ablation disentangles these roles: the taxonomy enables interpretable strategy characterization, but detection operates through domain-adapted embeddings regardless of label correctness.

Domain-adaptive pretraining. Continued pretraining on domain text yields consistent gains (Gururangan et al., 2020), as shown in biomedicine (Lee et al., 2020) and scientific text (Beltagy et al., 2019), extending ULMFiT (Howard and Ruder, 2018); yet this has not been disentangled for safety-critical generalization. Intermediate-task transfer (STILTs (Phang et al., 2018)) and multi-task learning (Caruana, 1997; Liu et al., 2019a) confound domain exposure with supervised signal. Our MLM control confirms pretraining alone closes the OOD gap. Dataset artifact analysis (Gururangan et al., 2018; Niven and Kao, 2019) motivates the TF-IDF baseline, adversarial NLP (Jin et al., 2020; Li et al., 2020; Nasr et al., 2025; Andriushchenko et al., 2025) informs the perturbation protocol, and probing (Conneau et al., 2018; Hewitt and Manning, 2019) underpins CKA analysis.

3 Method

Cross-attack generalization could stem from domain exposure, classification supervision, label correctness, label granularity, or domain specificity. Disentangling these factors demands an architecture where exactly one component varies across conditions. DisPeD separates encoder adaptation (Stage 1) from temporal classification (Stage 2). Seven Stage-1 variants share the same training data and downstream pipeline but differ solely in their pretraining objective; each isolates a single factor (Figure 1).

3.1 Problem Formulation

A conversation $C = \{u_1, a_1, \ldots, u_N, a_N\}$ alternates user turns u_i and assistant responses a_i. Given a prefix of K user turns, the model predicts $y \in \{0, 1\}$ (jailbreak vs. benign). We restrict input to user turns because adversarial intent resides in the attacker's messages. This formulation supports early detection from partial prefixes.

3.2 Stage 1: Turn-Level Encoder Adaptation

Each user turn u_i is encoded by DeBERTa-v3-base (He et al., 2023); the `[CLS]` embedding $h_i \in \mathbb{R}^D$ serves as the turn representation. Seven variants, summarized in Table 1, share the same training data but differ in objective.

The **9-class** variant uses two classification heads: 9-class persuasion strategy classification (L_{strategy}) and binary jailbreak intent (L_{intent}), with combined

Table 1: Stage-1 encoder variants. All operate on the same training conversations except Wikipedia MLM (matched token count on Wikipedia text).

Variant	Objective	Tests whether...
9-class	9-class persuasion + binary intent	full model
Binary	binary persuasion + binary intent	granularity matters
Scrambled	random 9-class + binary intent	label correctness matters
MLM-only	15% token masking	supervision is necessary
Topic	5-class topic classif.	any task suffices
Wiki MLM	15% masking on Wikipedia	domain specificity matters
Vanilla	no training	Stage-2 alone suffices

loss $L = L_{\text{strategy}} + \alpha \cdot L_{\text{intent}}$ ($\alpha = 0.3$, selected from $\{0, 0.3, 0.5, 2.0\}$; all values yield F1 = 1.0 on IID; Appendix X). The taxonomy comprises eight persuasion strategies (rapport building, authority appeal, emotional manipulation, logical reframing, role assignment, gradual escalation, obfuscation, direct request) plus a none category, grounded in PAP (Zeng et al., 2024b) and Cialdini's principles (Cialdini, 2001). Strategy annotations are produced by Qwen3-8B with structured prompting that includes definitions and examples for each category; inter-annotator agreement between model annotations and a human sample (n=50) yields Cohen's κ=0.78.

The **binary** variant collapses the 9-class taxonomy into persuasive vs. non-persuasive, testing whether label granularity matters for detection or robustness. The **scrambled** variant randomly permutes the 9-class labels across turns using a fixed random permutation (seed 42), preserving classification structure (cross-entropy loss, 9-class output space) while destroying label semantics. This isolates whether the classification objective provides representational benefits independent of label correctness. Robustness is verified across additional random permutations with consistent results (Appendix M). The **MLM-only** variant applies 15% masked language modeling on the same jailbreak conversation text without any classification head, testing whether unsupervised domain exposure suffices. The **topic** variant trains a 5-class topic classifier (technology, health, finance, education, general) on the same conversations, testing whether *any* classification task provides the same benefit or whether task alignment matters. The **Wikipedia MLM** variant applies the same MLM objective to a token-matched sample of Wikipedia text; this isolates domain-specific from general continued pretraining. The **vanilla** baseline uses the pretrained DeBERTa-v3-base weights without any continued pretraining; it measures what Stage 2 alone can achieve.

3.3 Stage 2: Conversation-Level Classification

Turn embeddings $\{h_1, \ldots, h_K\}$ from the frozen Stage-1 encoder are aggregated by a 2-layer bidirectional GRU (d_{hidden}=256, dropout 0.3): $\bar{h} = [\overrightarrow{h_K}; \overleftarrow{h_1}] \in \mathbb{R}^{512}$. A two-layer MLP ($512 \to 256 \to 2$, ReLU activation) maps $\bar{h}$ to $\hat{y} \in \{0, 1\}$. The Stage-1 encoder is frozen during Stage 2 training; only the BiGRU and MLP parameters are updated. This frozen-encoder design ensures ablation fairness: all seven Stage-1 variants are evaluated under identical downstream conditions, isolating the effect of Stage-1 representation quality from Stage-2 classifier capacity. Three-seed evaluation confirms that mean pooling matches BiGRU performance (Section 4.3). Cross-architecture validation on RoBERTa-base and -large (Section 4.4) yields thirteen configurations across three scales.

4 Experiments

Terminology. *Cross-attack OOD*: evaluation on attack families unseen during training; benign distribution remains IID throughout (OOD reflects attack-side shift only). K: prefix length (number of conversation turns retained). *DAPT*: all Stage-1 adaptations; we extend Gururangan et al. (2020) to supervised variants. *Evidence hierarchy* (on synthetic data): vanilla $<$ MLM $<$ classification (Section 4.3).

4.1 Setup

Data. All synthetic data is generated by Qwen3-8B (Yang et al., 2025). The IID dataset contains 250 Crescendo (Russinovich et al., 2025) jailbreak and 250 topic-anchored benign conversations, split into training (350), validation (74), and test (76) sets with balanced class ratios (e.g., test = 38 jailbreak + 38 benign). Stage 1 (DAPT) uses training + validation conversations; Stage 2 (BiGRU) trains on training, early-stops on validation, and evaluates on test. OOD evaluation pairs attack-specific jailbreak conversations with the 38 IID-test benign conversations: 80 FITD (Wang et al., 2024) (non-discriminative; Section 4.3) + 38 benign = 118 total,

80 Deceptive Delight (DD) (Palo Alto Networks Unit 42, 2024) + 38 benign = 118 total, and 80 ActorAttack (Ren et al., 2025) + 38 benign = 118 total. No overlap exists in attack templates across families; benign is IID (same generation pipeline). A paraphrased version of the IID test set (BERTScore 0.92, lexical overlap 0.20) serves as the adversarial robustness benchmark (Section 4.7). For human data validation, we use ToxicChat (Lin et al., 2023) (204 jailbreak + 204 benign from real LMSYS user interactions), entirely independent of Qwen3-8B generation.

Models. Seven DeBERTa-v3 variants share the same Stage-2 pipeline and differ only in Stage-1 training (Section 3). Our vanilla DeBERTa+BiGRU configuration is architecturally equivalent to DeepContext (Albrethsen et al., 2026) and differs only in encoder choice (DeBERTa-v3-base vs. BERT); it serves as our primary baseline. For cross-encoder validation, we replicate vanilla, MLM, and 9-class variants on RoBERTa-base (Liu et al., 2019b) (Section 4.4). We also compare TF-IDF LR, LLM-as-judge (Appendix A), and Llama Guard 3 8B (Meta AI, 2024) (zero-shot; evaluates last-response safety given conversation context), which detects only 1.8% of multi-turn jailbreaks (5/278 pooled recall; Table 3).

Metrics. F1 for binary jailbreak detection at prefix K=2 and full conversations (full per-K results in Appendix B). OOD results report 3-seed mean±std (seeds 42, 123, 456) for all variants. For key pairwise comparisons, we report BCa bootstrap 95% confidence intervals (10,000 resamples over individual predictions pooled across 3 seeds; this assumes cross-seed independence and is exploratory) and paired permutation tests (10,000 permutations). Seed-level F1 values (Table 38, Appendix Z.1) confirm all pairwise conclusions hold at every seed. All standard deviations are computed over 3 random seeds using the population formula (n divisor); Bessel-corrected estimates ($n-1$ divisor) are ≈22% larger but do not alter any significance conclusions.

Training. Stage-1: learning rate 2×10^{-5}, batch 16, 4 epochs, MLM masking 15%. Stage-2: two-layer MLP, 20 epochs.

4.2 Cross-Attack Generalization

Vanilla encoders collapse under cross-attack transfer; domain-adapted variants recover near-perfect detection (Table 2, Figure 2). DD exhibits high lexical separability (TF-IDF F1 = 0.961); ActorAttack, where TF-IDF collapses to 0.261, serves as the primary stress test for beyond-lexical generalization. On OOD DD, vanilla DeBERTa collapses to F1 = 0.312±0.026 and worsens with more turns (0.628 at K=1 to 0.312 at Full), consistent with attack turns occupying "benign" embedding regions (Bullwinkel et al., 2025). MLM achieves mean F1 ≥ 0.97 at all prefixes $K \leq 5$ without any classification labels; domain exposure alone is sufficient. At full conversation length, MLM drops to 0.914±0.115 (seed 456 recall = 0.64), while the 9-class model maintains 0.997±0.005. TF-IDF LR reaches F1 = 0.961 on full DD; DD is fully solvable by lexical features. On strategy-diverse ActorAttack, TF-IDF collapses to F1 = 0.261 (precision = 1.0, recall = 0.15), while 9-class maintains 0.972±0.026: a 71 pp gap. Lexical methods cannot capture multi-persona role-playing attack structure. This establishes a "difficulty threshold" for lexical methods parallel to that of vanilla encoders: both succeed on DD and fail on ActorAttack (Section 7). Seed-level results confirm consistent separation (Table 38); exploratory bootstrap CIs (pooled across seeds assuming independence) yield non-overlapping intervals (9-class DD Full [0.943, 1.000] vs. vanilla [0.202, 0.341]; permutation $p < 0.0001$). FITD is non-discriminative (all variants F1 ≥ 0.990; Section 7).

Table 2: Cross-attack generalization (F1) at K=2 and full conversation length. IID: Crescendo; OOD: Deceptive Delight (DD). 3-seed mean±std where noted; full per-K in Appendix B.

	IID (Cresc.)		*OOD (DD)*	
Method	K=2	Full	K=2	Full
9-class	**1.000±.000**	**1.000±.000**	.981±.013	**.997±.005**
MLM-only‡	.974	1.000	**.993±.009**	.914±.115
Vanilla	.978±.016	1.000±.000	.479±.038	.312±.026
TF-IDF LR§	1.000	1.000	1.000	.961

‡MLM-only IID reports single seed (42); OOD reports 3-seed.
§TF-IDF is deterministic (no std).

Prefix-length analysis (Appendix Table 7) shows vanilla F1 drops from 0.628 at K=1 to 0.312 at full length. MLM achieves strong early detection (K=2: 0.993) but is unstable at full length (0.914±0.115); the 9-class model maintains 0.972–0.997 across all prefixes. Classification supervision thus stabilizes against distributional drift.

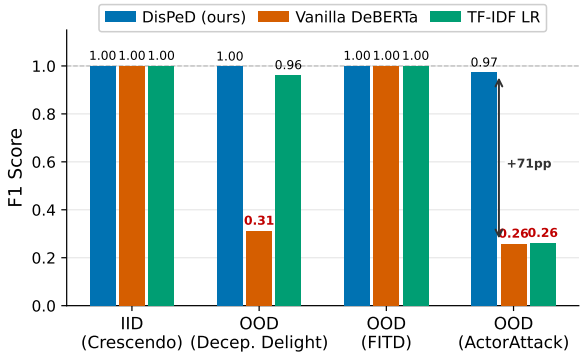


Figure 2: Cross-attack overview (F1 at full conversation length). DisPeD (9-class) maintains near-perfect detection across IID and both OOD families; vanilla collapses on OOD DD. FITD is non-discriminative for all methods.

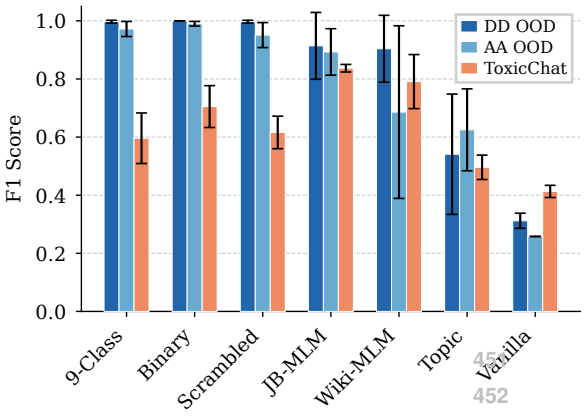


Figure 3: Evidence hierarchy across evaluation conditions. Classification-based DAPT outperforms MLM and vanilla on synthetic OOD attacks (DD, AA), but the hierarchy reverses on human-authored single-turn data (ToxicChat), where JB-MLM leads.

4.3 Decomposing the Detection Mechanism

A clear *evidence hierarchy* emerges across seven variants (Table 3), ordered by OOD detection at full conversation length on DD:

vanilla (0.312±0.026) < topic (0.541±0.207) ≪ wiki-MLM (0.904±0.115) ≈ jb-MLM (0.914±0.115) < scrambled (0.997±0.005) = 9-class (0.997±0.005) ≈ binary (1.000±0.000)

Domain exposure is the primary observed factor. MLM bridges ∼60 pp from vanilla (0.312±0.026 → 0.914±0.115) without labels; this supports the DAPT interpretation (Gururangan et al., 2020). TF-IDF collapse on ActorAttack (0.261 vs. 9-class 0.972) confirms that domain-adapted representations capture features beyond lexical overlap. All three classification-supervised variants converge to equivalent detection (F1 = 0.997–1.000); Mc-

Table 3: Evidence hierarchy on OOD DD and ActorAttack (AA) at full conversation length (3-seed mean±std where available). OOD F1 reflects attack-side generalization only; benign samples are IID from training distribution (benign-side shift evaluated in Section 4.9). Bold indicates best per column. All variants share the same Stage-2 pipeline. Full per-prefix results in Appendix Tables 7 and 8. Bottom rows: full dual-OOD evaluation with WildChat benign replacing IID benign (Appendix P.1).

Variant	DD-Full	AA-Full	DD-K=2
9-class (full model)	.997±.005	.972±.026	.981±.013
Binary (2-class)	**1.000±.000**	**.990±.008**	.994±.009
Scrambled labels	.997±.005	.951±.043	.981±.026
MLM-only‡ (no labels)	.914±.115	.893±.080	.993±.009
Mean pooling*	.991±.013	.962±.015	**1.000**
Topic classif. (5-class)	.541±.207	.625±.141	.593±.166
Wikipedia MLM	.904±.115	.686±.297	.911±.073
Vanilla DeBERTa	.312±.026	.258±.000	.479±.038
TF-IDF LR†	.961	.261	**1.000**
Llama Guard 3¶	.025	.000	–
Dual-OOD (WildChat benign; Section 4.9)			
9-class (no adapt)	.855	.842	–
9-class (+10-shot)	.994	.981	–

*9-class encoder with mean pooling replacing BiGRU. †Deterministic (no std). ‡MLM-only IID (Table 2) is single-seed (42); all OOD values are 3-seed. ¶Zero-shot; no domain training.

Nemar's ($p > 0.13$) and paired permutation tests ($p \geq 0.40$) provide descriptive evidence suggesting equivalence. Extended to $n = 10$ seeds, TOST (margin = 5 pp) confirms equivalence on DD ($p < 10^{-8}$, $d = -0.14$; Table 16) and FITD ($p < 10^{-8}$, $d \approx 0$). An observed difficulty gradient emerges: template attacks show full equivalence ($|d| \leq 0.14$), while strategy-diverse ActorAttack retains moderate label sensitivity ($d = 0.44$, $p = 0.183$). Label quality and granularity are descriptively non-factors for detection. Classification supervision is associated with descriptively 3–23× lower cross-seed variance over MLM on synthetic OOD (std 0.005 vs. 0.115 on DD; observed range across 3 seeds; formal variance ratio tests are not applicable at n=3; cross-seed ensemble yields marginal additional gains, Appendix Z). This stability benefit does not transfer to human-authored data, where classification-supervised variants exhibit the highest variance and pure MLM the lowest (Section 4.6). Appendix G provides t-SNE visualizations confirming that domain-adapted variants produce tighter, more separable embedding clusters than the vanilla baseline.

Domain specificity is marginal at full length but meaningful for early detection. Wikipedia

MLM (0.904±0.115) nearly matches jailbreak-domain MLM at full length (1 pp gap), though jailbreak-domain text provides an 8 pp advantage at early prefixes (K=2: 0.993 vs. 0.911). Perturbation analysis (Section 4.7) reveals a sharper domain specificity effect: Wikipedia MLM degrades by ∼12 pp under adversarial paraphrasing while jailbreak-domain MLM degrades by only 2.6 pp.

Misaligned supervision destabilizes generalization. Topic classification yields DD-Full F1 = 0.541±0.207; seed 123 falls *below* vanilla (0.252 vs. 0.312). Despite using the same jailbreak-domain training data, topic classification performs worse than Wikipedia MLM (0.904±0.115), which uses out-of-domain text with no labels. A misaligned objective forces the encoder to allocate capacity to irrelevant features and actively destabilizes representations that MLM would preserve.

Evidence hierarchy on strategy-diverse attacks. ActorAttack (Ren et al., 2025) uses multi-persona steering across longer conversation structures and provides a more challenging OOD benchmark than DD. The evidence hierarchy at full conversation length is:

TF-IDF (0.261) ≈ vanilla (0.258±0.000) < topic (0.625±0.141) < wiki-MLM (0.686±0.297) < jb-MLM (0.893±0.080) < scrambled (0.951±0.043) ≈ 9-class (0.972±0.026) ≈ binary (0.990±0.008)

TF-IDF collapses from 0.961 (DD) to 0.261 (ActorAttack, precision = 1.0, recall = 0.15). This gap shows that domain-adapted representations capture features beyond bag-of-words. All classification-supervised variants converge to equivalent performance (0.951–0.990, max gap 3.9 pp); this extends the DD finding. DAPT provides ∼71 pp over both vanilla (0.258) and TF-IDF (0.261). Full per-prefix results and per-prefix error analysis in Appendix C and C.1.

Temporal modeling. Mean pooling matches BiGRU on both DD and ActorAttack (F1 gap ≤ 1 pp, overlapping CIs; Appendix T). Turn-order shuffling produces negligible change ($|\Delta| < 1$ pp) across both DeBERTa-v3-base 9-class (DD $\Delta = 0.000$, AA $\Delta = -0.009$, IID $\Delta = 0.000$) and RoBERTa-large 9-class (DD $\Delta = +0.001$, AA $\Delta = -0.007$, IID $\Delta = 0.000$); detection relies on per-turn representations rather than sequential patterns. The equivalence of mean pooling and BiGRU suggests that per-turn representations encode sufficient discriminative signal, with temporal order providing negligible additional information. This challenges the assumption that multi-turn attacks require sequential modeling and suggests detection operates on per-turn features aggregated independently.

Table 4: Cross-encoder comparison on OOD Full (3-seed mean±std). DD = Deceptive Delight, AA = ActorAttack. Bold indicates best per encoder group.

Variant	DD	AA
DeBERTa-v3-base (184M)		
Vanilla	.312±.026	.258±.000
MLM	.914±.115	.893±.080
9-class	**.997±.005**	**.972±.026**
RoBERTa-base (125M)		
Vanilla	.981±.026	.466±.163
MLM	.987±.018	.730±.201
9-class	**.994±.005**	**.939±.046**
RoBERTa-large (355M)		
Vanilla	.927±.036	**1.000±.000**
MLM	.884±.065	.874±.014
9-class	**.987±.004**	.946±.057

4.4 Cross-Encoder Validation

To test whether the evidence hierarchy depends on encoder choice, we replicate vanilla, MLM, and 9-class on RoBERTa-base (Liu et al., 2019b) (Table 4). On DD, RoBERTa-base vanilla already achieves 0.981±0.026 (vs. DeBERTa 0.312±0.026), so DAPT adds only 1.3 pp. ActorAttack exposes this as DD's simplicity: RoBERTa-base vanilla collapses to 0.466±0.163, a 51.5 pp drop, while domain-adapted variants recover (9-class: 0.939±0.046).

RoBERTa-large (355M) reveals a capacity-DAPT interaction: its effect on ActorAttack reverses (vanilla 1.000 vs. 9-class 0.946), while smaller encoders require DAPT to bridge 47–71 pp. All continued pretraining reduces ActorAttack performance for the largest encoder via representation drift (Table 15), though classification DAPT partially compensates on DD (+6 pp).

Three encoder scales thus reveal a scale-dependent pattern for the *difficulty threshold*: DeBERTa-v3-base requires adaptation for all attacks, RoBERTa-base only for complex ones, and

RoBERTa-large for none (Section 5). Classification supervision provides a consistent stability benefit across both base encoders (cross-seed variance: 0.005–0.046 vs. 0.026–0.201). A BERT-base-uncased proxy (110M, matching DeepContext's encoder family) without adaptation achieves DD = 0.318±0.031 and AA = 0.275±0.027, comparable to vanilla DeBERTa-v3 (Appendix Q); cross-attack generalization appears primarily associated with DAPT rather than encoder architecture.

4.5 Mixed-LLM Training

All preceding experiments use single-source (Qwen3-8B) data. To test whether the evidence hierarchy transfers when training data mixes LLM sources, we train on 250 Qwen + 250 Llama-3.1-8B conversations (Table 13). Classification-based DAPT is the only variant that substantially outperforms vanilla: 9-class achieves DD OOD F1 = 0.818±0.096 (+53 pp over vanilla 0.291±0.065), while MLM drops to vanilla level (0.289±0.101). On strategy-diverse ActorAttack, ActorAttack detection degrades to chance level (9-class F1 = 0.504±0.107, indistinguishable from majority baseline ∼0.5 at $n = 3$). DD and FITD thus primarily validate the baseline rather than the method; ActorAttack, where lexical methods collapse (TF-IDF F1 = 0.261) and vanilla encoders fail (F1 ≤ 0.389 across conditions), provides the discriminative evaluation axis. An alternative explanation for the mixed-LLM performance degradation is per-source data reduction (250 vs. 350 conversations per source) rather than distribution shift per se. Disentangling these factors would require training on 350 conversations from each source independently, which we leave to future work. Mixed-LLM training confirms that classification supervision generalizes across LLM sources where unsupervised pretraining does not; full results in Appendix I.

4.6 Human Data Validation

Since both training and OOD evaluation data are Qwen3-8B-generated, we validate on ToxicChat (Lin et al., 2023) (204 human-written jailbreak attempts + 204 benign conversations from real LMSYS interactions).

All DAPT variants exceed vanilla by 18–42 pp (Table 5): the adapted-vs.-vanilla gap is not a Qwen3-8B generation artifact, and domain exposure on synthetic jailbreak text transfers to human-written attacks.

Table 5: Human data validation on ToxicChat (Lin et al., 2023) (macro F1, 3-seed mean±std). Bold indicates best. Single-turn data from real LMSYS user interactions, independent of Qwen3-8B.

Variant	Macro F1
JB-MLM	**.837±.013**
Wiki-MLM	.791±.093
Binary	.705±.072
Scrambled	.616±.056
9-class	.596±.087
Topic	.496±.042
Vanilla	.413±.021

The ordering reverses: JB-MLM leads (0.837±0.013) while 9-class ranks lower (0.596±0.087; $p < 0.001$, non-overlapping CIs). ToxicChat is single-turn, so the classification head provides no advantage; validating the multi-turn hierarchy requires multi-turn human data.

4.7 Perturbation Analysis

Under adversarial paraphrasing (BERTScore 0.92), classification-supervised variants achieve zero degradation on IID and OOD DD (Table 10) while non-adapted models degrade by ∼12 pp ($p < 0.001$); on ActorAttack, DAPT variants degrade ($\Delta = -20.4$ pp) but maintain 52 pp over vanilla (Appendix D).

4.8 Strategy Trajectory Analysis

We operationally distinguish *template attacks* (fixed prompt template with variable payload; e.g., Crescendo, Deceptive Delight) from *structurally complex attacks* (multi-turn strategy adaptation with dynamic context exploitation; e.g., ActorAttack). A quantitative proxy is per-turn strategy transition rate: template attacks show rate < 0.1 (consistent strategy), while complex attacks show rate > 0.3 (frequent strategy shifts).

The 9-class taxonomy provides interpretable characterization of known attacks (κ=0.78, n=50; Appendix E) but degrades to near-random on OOD (κ=0.126; Appendix Y). Its value is twofold: as ablation infrastructure (scrambled labels produce equivalent detection) and as an interpretability tool for known attacks; detection operates through domain-adapted `[CLS]` embeddings, not strategy accuracy.

4.9 Benign Distribution Shift

All preceding OOD results use IID benign data. On WildChat benign conversations, 9-class FPR rises

to 68.1%; cross-LLM error decomposition confirms false positives on unfamiliar benign text drive the collapse (Appendix P, H). Few-shot adaptation fine-tunes the Stage-2 BiGRU classifier (Stage-1 encoder frozen) on N target-domain benign conversations using BCE loss for 5 epochs; adaptation and evaluation sets are disjoint. This resolves benign-side shift without target-domain attack data: 10 WildChat conversations reduce FPR to 0%, and 25 OASST2 conversations reduce FPR to 1.0% (Appendix P, P); dual-OOD F1 drops 12–14 pp without adaptation but recovers within 0.3 pp (Appendix P.1).

5 Discussion

Discriminative compression and label equivalence. Three observations argue against a topic shortcut: unstable topic OOD (F1 = 0.541±0.207), scrambled labels within 3.9 pp, and label-free MLM detection (F1 ≥ 0.97). Classification DAPT compresses features into few dimensions (participation ratio 3.7–9.4 vs. 25–32; Fisher 2.6–19.7 vs. <0.001; Appendix S.1), consistent with Zhang et al. (2021): domain-relevant partitioning appears necessary (topic DAPT fails; Appendix M) and compression localizes to 3 SVD dimensions in layers 8–12 (Appendix S.1). A probe over these dimensions confirms sufficiency; removal degrades DD by 17 pp while FITD is unaffected (partial necessity; Appendix S.1). This aligns with concurrent findings that internal LLM layers encode detectable jailbreak patterns (Kadali and Papalexakis, 2025, 2026).

Cross-condition transfer and boundaries. ToxicChat (+18–42 pp; Table 5) and cross-LLM experiments (Appendix H) confirm transfer beyond single-LLM artifacts; mixed-LLM 9-class maintains 0.818 while MLM collapses, suggesting classification extracts style-invariant features. Classification supervision is associated with lower variance on synthetic OOD (descriptively 3–23×, 3 seeds) but this reverses on ToxicChat (std = 0.087 vs. 0.013); frozen probing localizes MLM instability to Stage-2 (Appendix S). Three encoders (125M–355M) differ in pretraining; same-architecture scaling is needed. At larger scales (355M), DAPT can be counterproductive: RoBERTa-large vanilla achieves AA F1 = 1.000 while 9-class drops to 0.946. WildChat FPR rises from 48.1% to 68.1% under 9-class DAPT, confirming that attack-side generalization amplifies benign distribution sensitivity.

An alternative interpretation of the low-dimensional discriminative subspace (Section 4.3) is overfitting or representation collapse rather than beneficial feature selection. The 68.1% FPR on out-of-distribution benign data provides partial evidence for this interpretation: classification DAPT may compress representations in ways that sacrifice generalization to unseen benign distributions while improving attack-side separation.

Length as a potential confound. Turn count differs between attack families (DD: 6–10 turns, AA: 3–8) and IID benign (6–14), raising the concern that detection may exploit length shortcuts. Length-matched evaluation on the full OOD test set (N=155, rather than the balanced subset in Table 2) over the overlapping turn-count range shows negligible change: DD F1 shifts by −0.8 pp (range-matched) to +2.5 pp (bucket-matched), AA by +1.5 pp to +0.9 pp, confirming that detection operates on per-turn content rather than conversation length. The −57.9 pp ActorAttack collapse under zero-padding (Appendix U) is consistent with signal density dilution—padding introduces uninformative turns that reduce the proportion of attack-bearing content—rather than length shortcut removal.

6 Conclusion

We decompose multi-turn jailbreak generalization into two orthogonal axes. On synthetic multi-turn data at base model scale (125M–184M), classification-based domain adaptation recovers out-of-distribution detection (F1 = 0.97–1.0 vs. vanilla 0.26–0.31) while unsupervised pretraining collapses; template attacks remain robust across conditions ($|d| \leq 0.14$) but strategy-diverse attacks degrade under mixed-LLM training (F1 = 0.504). Orthogonal to attack-side generalization, few-shot benign adaptation reduces false-positive rates from 58–68% to 0–1.5% with as few as 10 conversations, at 24.3× inference speedup over LLM-as-judge.

On human-authored single-turn data (ToxicChat), this hierarchy reverses: MLM outperforms classification (0.837 vs 0.596, a 24.1 pp gap). Classification DAPT's advantage is condition-specific (multi-turn, synthetic) rather than universal.

These findings have boundaries. At larger scales (355M), DAPT can be counterproductive (RoBERTa-large vanilla AA = 1.000 vs. 9-class 0.946). Mixed-LLM diversity degrades strategy-

diverse attack detection to near chance, and padding sensitivity persists across configurations.

The ablation framework contributes a reusable methodology for safety classification research: four adaptation axes (domain exposure, supervision type, label quality, granularity) at multiple scales provide a template for diagnosing when and why domain adaptation succeeds or fails.

7 Limitations

- **Synthetic data only.** All findings derive from synthetic conversations (single LLM, Qwen3-8B; 350 training, 76–118 test, 3 seeds). The evidence hierarchy reverses on human single-turn data (ToxicChat; Section 5), and no human-generated multi-turn jailbreak dataset exists.
- **IID benign upper bound.** OOD F1 figures assume IID benign data; dual-OOD evaluation drops 12–14 pp without adaptation (Table 2). Few-shot benign adaptation recovers performance but is validated on only two platforms ($n \leq 49$).
- **Scope.** Only persuasion-mediated multi-turn attacks are addressed; single-turn token-level attacks and non-text modalities are out of scope.

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A LLM-as-Judge Baseline

Table 6 reports LLM-as-judge performance on IID and OOD data. On IID Crescendo, GPT-5.5 achieves precision of 1.0 at all turn prefixes but recall ranges from only 0.105 to 0.290, resulting in F1 = 0.383 on full conversations; 30–47% of responses are unparseable. Qwen3-8B performs worse (IID F1 = 0.100). On OOD data, zero-shot Qwen3-8B achieves macro F1 = 0.244 on both DD and ActorAttack by predicting nearly all conversations as benign. Even with 3-shot prompting, Qwen3-8B reaches only 0.566 (DD) and 0.652 (AA), while DisPeD maintains macro F1 $\geq$ 0.972. The poor performance of content-level reasoning on unseen attack families underscores the necessity of trained classifiers with domain-adapted representations.

Table 6: LLM-as-judge jailbreak detection. IID: binary F1 (Crescendo test set); OOD: macro F1 (DD = Deceptive Delight, AA = ActorAttack). GPT-5.5 F1 values exclude unparseable responses and represent upper bounds.

Method	DD	AA	IID	Unparse.
Zero-shot				
GPT-5.5	–	–	.383	30–47%
Qwen3-8B	.244	.244	.100	0%
3-shot				
Qwen3-8B	.566	.652	–	0%
Trained classifier				
9-class (ours)	**.994**	**.972**	**1.000**	0%

B Full Cross-Attack Results by Prefix

C ActorAttack Full Results

Table 8 reveals three distinct prefix-length patterns. The 9-class model improves monotonically from K=1 to Full (+20 pp), consistent with cumulative evidence accumulation across turns. Vanilla DeBERTa *decreases* with more turns (0.337 at K=1 to 0.258 at Full), suggesting that additional attack turns push representations further into benign embedding regions. MLM peaks at K=3 (0.945) then declines to 0.893 at Full, indicating that unsupervised domain exposure captures early-turn patterns but is unstable under extended sequences.

C.1 Error Analysis

Per-sample majority vote across 3 seeds yields 98.3% accuracy on ActorAttack OOD (2 false negatives, 0 false positives). No sample is misclassified by all three seeds; all 7 seed-specific errors are false negatives concentrated in shorter conversations (3–6 turns). On these 7 error samples, vanilla achieves 0/3 correct votes per sample, while 9-class majority vote correctly classifies 5/7, confirming that domain adaptation provides robust per-sample

Table 7: Full cross-attack generalization results (F1) at all prefix lengths. 3-seed mean±std where available. "–" indicates insufficient conversation length for that prefix (FITD conversations contain 2–3 turns).

Method	K=1	K=2	K=3	K=5	Full
IID (Crescendo)					
9-class (ours)	1.000±.000	1.000±.000	1.000±.000	1.000±.000	1.000±.000
MLM-only	.960	.974	.987	1.000	1.000
Vanilla DeBERTa	.968±.033	.978±.016	.987±.000	.982±.014	1.000±.000
TF-IDF LR	1.000	1.000	1.000	1.000	1.000
OOD (Deceptive Delight)					
9-class (ours)	.990±.013	.981±.013	.972±.026	.997±.005	.994±.005
Binary	.994±.009	.994±.009	.984±.008	1.000±.000	1.000±.000
Scrambled	.997±.005	.981±.026	.975±.029	.997±.005	.994±.005
MLM-only	.973±.019	.993±.009	.997±.005	.997±.005	.914±.115
Vanilla DeBERTa	.628±.038	.479±.038	.389±.027	.351±.030	.312±.026
TF-IDF LR	1.000	1.000	.994	1.000	.961
OOD (FITD)					
9-class (ours)	1.000±.000	1.000±.000	–	–	1.000±.000
Vanilla DeBERTa	.954±.039	.952±.061	–	–	.994±.009
TF-IDF LR	1.000	–	–	–	1.000

Table 8: Full ActorAttack OOD results (F1) at all prefix lengths. 3-seed mean±std (Binary per-K reports n=2; Full is 3-seed).

Method	K=1	K=2	K=3	K=5	Full
DeBERTa-v3					
Binary	.788±.020	**.935±.018**	**.976±.005**	**.995±.005**	**.990±.008**
9-class	.773±.041	.855±.048	.921±.046	.966±.028	.972±.026
Scrambled	.652±.034	.804±.053	.854±.046	.910±.052	.951±.043
MLM-only	.736±.150	.909±.058	.945±.039	.929±.033	.893±.080
Wiki MLM	.656±.224	.672±.252	.693±.280	.689±.285	.686±.297
Topic	.513±.077	.580±.109	.565±.100	.615±.111	.625±.141
Vanilla	.337±.049	.306±.026	.284±.011	.262±.006	.258±.000
TF-IDF LR	.633	.730	.571	.242	.261
RoBERTa					
9-class	**.738±.033**	.805±.033	.884±.066	.925±.066	.939±.046
MLM	.633±.133	.665±.150	.713±.174	.719±.184	.730±.201
Vanilla	.410±.126	.425±.143	.432±.144	.466±.144	.466±.163

coverage even where individual seeds fail.

D Perturbation Robustness

Table 9 shows perturbation behavior on IID data. Classification-supervised variants (9-class, binary, scrambled) achieve zero degradation at full conversation length regardless of label correctness. Models without jailbreak-domain exposure (Wikipedia MLM, vanilla) degrade by $\sim$12 pp. McNemar's test finds no significant pairwise differences among classification-supervised variants (all $p > 0.13$), though with $\leq$7 discordant pairs, statistical power is limited.

On OOD DD, the IID pattern replicates: classification-supervised variants show zero degradation while McNemar's test confirms 9-class vs. vanilla significance ($p < 0.001$, 49–78 discordant pairs). On OOD ActorAttack, domain-adapted variants degrade substantially (9-class $\Delta = +20.4 \pm 13.5$ pp, MLM -35.3 pp), but perturbation narrows rather than closes the adapted vs. non-adapted gap: 9-class perturbed F1 = 0.768 vs. vanilla 0.248 (McNemar $p < 0.001$). Domain specificity also manifests: Wikipedia MLM degrades by $\sim$12 pp while jailbreak-domain MLM degrades by only 2.6 pp.

E Strategy Trajectory Analysis

Figure 3 shows transition count matrices for three attack families. Crescendo conversations exhibit broad transition coverage across all nine strategy categories (1,818 total transitions), with the highest concentration in Role Assignment self-transitions (545); this reflects its gradual escalation design.

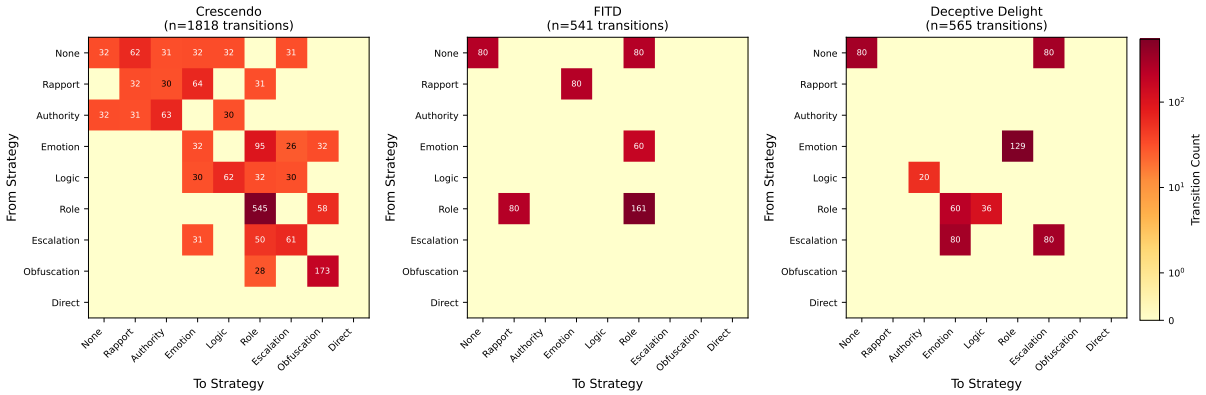


Figure 4: Persuasion strategy transition count matrices for three attack families, based on ground-truth intended strategy labels. Strategy labels validated on n=50 samples with a single annotator (κ=0.78); fine-grained trajectory patterns should be interpreted with this limitation. Crescendo exhibits broad coverage across all strategy categories, while FITD and DD concentrate transitions in smaller subsets. Row i, column j counts transitions from strategy i at turn t to strategy j at turn $t+1$.

Table 9: Perturbation behavior under adversarial paraphrasing on IID Crescendo (n=76). Δ: F1 degradation (pp). Classification-supervised variants achieve zero degradation at full length; this pattern does not generalize to ActorAttack.

Method	Δ Full	Δ K=1	Δ K=2	Δ K=3
9-class (ours)	**0.0**	**−2.6**	**0.0**	**0.0**
Scrambled	**0.0**	−5.3	−2.6	**0.0**
Binary	**0.0**	−9.2	−2.6	**0.0**
TF-IDF LR	−1.3	−9.6	−4.1	−1.3
MLM-only	−2.6	−9.2	−4.0	−4.0
Topic (seed 42)	−4.0	−2.7	−2.7	**0.0**
Wikipedia MLM	−12.1	−12.4	−15.4	−17.9
Vanilla DeBERTa	−11.8	−14.6	−12.1	−13.8

Table 10: Perturbation robustness on OOD Deceptive Delight (n=118). Δ: F1 change (pp) from clean to paraphrased. 3-seed mean±std where available.

Variant	Clean Full	Pert. Full	Δ Full	Δ K=1
9-class	.994±.005	1.000±.000	+0.6	+0.6
Binary	1.000±.000	1.000±.000	0.0	+0.6
Scrambled	.994±.005	1.000±.000	+0.6	−0.3
JB MLM	.914±.115	.941	+2.7	0.0
Wiki MLM*	.904±.115	.872	−3.3	−3.2
Vanilla†	.312±.026	.339±.055	+2.7	−6.8±1.1

*Wiki MLM degradation is seed-driven: seed 42 accounts for −9.8 pp; other seeds show 0.0 pp change.
†Vanilla Δ Full = +2.7 pp reflects floor-effect noise (baseline 0.312, std 0.055), not meaningful improvement.

FITD conversations show a markedly sparser pattern (541 transitions), with Role→Obfuscation (161) as the dominant transition, consistent with FITD's commitment-consistency exploitation (Cialdini, 2001). DD conversations span 5 active source strategies (565 transitions), with Emotion→Obfuscation (129) as the dominant transition, aligning with DD's design of embedding harmful content within emotionally engaging contexts (Palo Alto Networks Unit 42, 2024).

F Centered CKA Heatmaps

G Embedding Space Visualization

Figure 5 visualizes mean-pooled `[CLS]` embeddings via t-SNE for three representative variants (Vanilla, JB-MLM, 9-Class) on DD OOD and ActorAttack. Vanilla embeddings show dispersed jailbreak clusters with ambiguous decision boundaries, while domain-adapted variants (JB-MLM, 9-Class) produce tighter, more separable clusters. JB-MLM achieves the clearest bimodal separation on both evaluation sets, consistent with its superior detection performance.

H Cross-LLM Validation

Table 11 shows three inference conditions. When both jailbreak and benign text come from Llama (row 2), all variants collapse (F1 = 0.26–0.43), but this conflates two sources of distribution shift: unfamiliar jailbreak patterns and unfamiliar benign text. Row 3 isolates jailbreak transferability by pairing Llama jailbreaks with the original Qwen benign distribution. Under this condition, 9-class recovers to F1 = 1.000 on DD and 0.801 on ActorAttack, and JB-MLM reaches 0.903 on ActorAttack. The contrast between rows 2 and 3 shows that the dominant failure mode in row 2 is benign distribution mismatch (false positives from unfamiliar

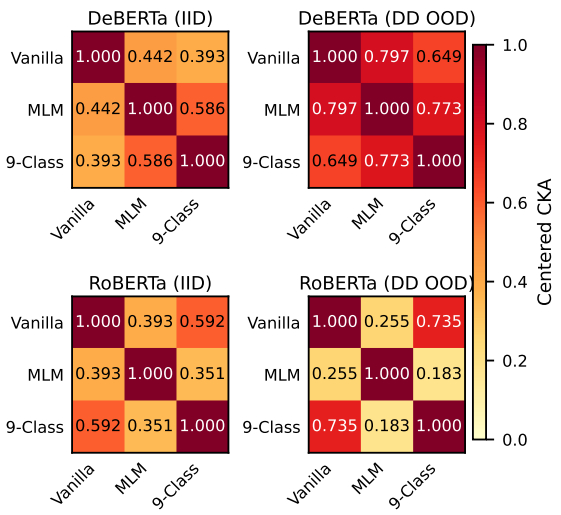


Figure 5: Centered CKA similarity (Kornblith et al., 2019) between encoder variants on IID and OOD data for DeBERTa-v3 and RoBERTa. Low off-diagonal values (vanilla vs. adapted variants) confirm that continued pretraining substantially reshapes representations; the MLM vs. 9-class block shows distinct geometries (CKA ≤ 0.2 on RoBERTa OOD).

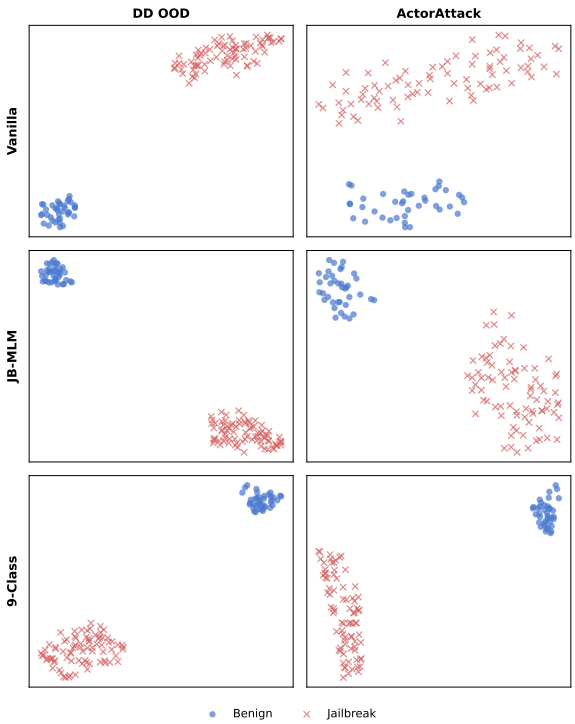


Figure 6: t-SNE visualization of mean-pooled `[CLS]` embeddings (perplexity=30) for Vanilla, JB-MLM, and 9-Class variants on DD OOD and ActorAttack (118 samples each; blue=benign, red=jailbreak).

Table 11: Cross-LLM validation (F1, full conversation). All models are trained on Qwen3-8B data; conditions vary the source of jailbreak (JB) and benign (ben.) conversations at inference. Values are evaluated under a matched 50-sample subset protocol and differ from Table 3, which uses the full evaluation set.

	DD Full			*AA Full*		
Condition	Van.	MLM	9-cl.	Van.	MLM	9-cl.
Qwen JB + Qwen ben.	.312	.914	.997	.258	.893	.972
Llama JB + Llama ben.	.261	.282	.433	.261	.368	.921
Llama JB + Qwen ben.	.302	.778*	1.000	.302	.903	.801

*High seed variance (std = 0.306): seed 456 outlier (F1 = 0.345); remaining two seeds achieve F1 $\geq$ 0.97.

Table 12: Error decomposition for DD Full cross-LLM conditions (matched 50-sample subsets). TP rate is identical across benign conditions; the F1 collapse in the Llama-only condition (Table 11, row 2) is driven entirely by false positives on unfamiliar benign text.

Metric	Van.	MLM	9-cl.
TP rate (Llama JB)	0.0%	68.0%	100.0%
FP rate (Llama ben.)	29.3%	88.0%	90.0%
FP rate (Qwen ben.)	0.0%	0.9%	0.0%

benign text), not inability to recognize cross-LLM jailbreak patterns. Vanilla remains at 0.26–0.31 across all conditions, further confirming that encoders without DAPT lack OOD detection capability regardless of the LLM source. The residual degradation on ActorAttack (0.801 vs. 0.972) reflects genuine cross-LLM variation in structurally complex attack patterns.

Table 12 decomposes DD Full F1 into true positive and false positive rates. TP rates are identical regardless of the benign source: domain-adapted models detect Llama-generated jailbreaks at the same rate whether evaluated against Qwen or Llama benign text. The F1 collapse in the full Llama condition (Table 11, row 2) is driven entirely by false positives on unfamiliar benign text (FP rate: 88–90% on Llama vs. 0–0.9% on Qwen for adapted variants). The model implicitly treats non-Qwen text style as a jailbreak indicator, a spurious correlation from single-LLM training data. This failure reflects a data diversity limitation, not a methodological one; mixed-LLM training (Table 13) partially addresses this confound.

I Mixed-LLM Training Results

Table 13 reports detection performance when training on a mixed corpus of Qwen3-8B and Llama-3.1-8B conversations. Classification-based DAPT (9-class) maintains the strongest DD OOD performance (0.818±0.096), outperforming both vanilla

Table 13: Mixed-LLM training (250 Qwen + 250 Llama conversations, 3-seed mean±std where available). ToxicChat evaluated on balanced set (408 conversations).

Variant	Mixed	DD OOD	AA OOD	FITD OOD	ToxicChat
9-class	**.996±.006**	**.818±.096**	**.504±.107**	**.997±.005**	**.550±.052**
MLM	.681±.226	.289±.101	.258±.075	.679±.188	.256±.033
Vanilla	.802±.010	.291±.065	.389±.073	.867±.022	.556±.009

(0.291±0.065) and MLM (0.289±0.101) by 53 pp. ActorAttack remains challenging under mixed-LLM conditions (9-class F1 = 0.504±0.107), suggesting structurally complex attacks require larger or more diverse training data. ToxicChat performance shows no significant improvement over single-source training (0.550±0.052 vs. 0.596±0.087; Section 4.6), consistent with ToxicChat being zero-shot single-turn transfer rather than a standard OOD benchmark.

J Dataset Scale Validation

Table 14: Dataset scale validation (1,000 Qwen conversations, 3-seed mean±std). IID results omitted due to train/test overlap (71% of original test set present in expanded training data).

Variant	DD OOD	AA OOD
9-class	**.889±.041**	**.913±.041**
MLM	.338±.010	.351±.018
Vanilla	.326±.000	.508±.085

Table 14 validates robustness to dataset scale by doubling the training corpus from 500 to 1,000 conversations with new topic seeds. The hierarchy 9-class ≫ MLM ≈ vanilla holds at 2× scale: classification-based DAPT maintains 40–56 pp advantage over vanilla, while MLM provides no improvement over vanilla on either attack family (DD: 0.338 vs. 0.326; AA: 0.351 vs. 0.508).

K RoBERTa-large DAPT Learning Rate Control

Table 15: RoBERTa-large (355M) DAPT learning rate control on OOD Full (3-seed mean±std). Reducing the learning rate worsens both DD and AA, ruling out over-adaptation as the cause of the regression in Table 4.

Config	DD Full	AA Full
Vanilla (no DAPT)	.927±.036	1.000±.000
9-class, LR=2×10^{-5}	.987±.004	.946±.057
9-class, LR=1×10^{-5}	.948±.034	.781±.054

Reducing the DAPT learning rate from 2×10^{-5} to 1×10^{-5} worsens both DD and AA performance, which rules out over-adaptation as the cause of the regression observed in Table 4. An MLM-only variant (without classification) produces the same pattern (AA Full = 0.874±0.014); representation drift from continued pretraining, not classification-specific bias, is the underlying cause. Domain-adaptive fine-tuning narrows the representation space of a sufficiently large encoder and reduces its inherent OOD generalization ability.

L TOST Equivalence Sensitivity Analysis

Table 16: TOST equivalence test sensitivity to margin δ (scrambled vs. 9-class, 3-seed). p-values from two one-sided t-tests; $p < 0.05$ confirms equivalence within $\pm\delta$.

Dataset	δ=5 pp	δ=3 pp	δ=2 pp
DD OOD	.006 ✓	.016 ✓	.035 ✓
ActorAttack	.290	.428	.507

On DD, TOST confirms scrambled ≈ correct labels even at δ=2 pp (p=0.035), demonstrating that label correctness is not a detection factor; the DD equivalence conclusion is robust to margin choice. On ActorAttack, the mean gap is 2.1 pp but TOST is underpowered (SE = 0.045, n=3) to confirm formal equivalence at any tested margin; the variants are descriptively similar but formally inconclusive.

M Partition Robustness and Topic Failure Analysis

M.1 Random Binary Partition Robustness

Table 17: OOD F1 for 10 random binary partitions of the 9 strategy labels (each partition randomly assigns strategies to group A or B). All partitions use identical Stage-1 DAPT and Stage-2 GRU training; only the binary label mapping differs. Seed 42 (3 runs).

Partition	Group A	IID	DD OOD	AA OOD
0	{0,2,4,5,7}	1.000	1.000	1.000
1	{1,3,5,6,8}	1.000	1.000	1.000
2	{0,1,3,7,8}	1.000	1.000	1.000
3	{2,4,5,6,7}	1.000	1.000	1.000
4	{0,1,2,6,8}	1.000	1.000	1.000
5	{0,3,4,7,8}	1.000	1.000	1.000
6	{1,2,3,5,6}	1.000	1.000	1.000
7	{1,4,6,7}	1.000	1.000	0.636
8	{0,2,3,4,8}	1.000	1.000	1.000
9	{0,1,5,7,8}	1.000	1.000	1.000
Mean±Std		1.000±.000	1.000±.000	0.964±.109

All 10 partitions achieve perfect IID and DD OOD detection regardless of how the 9 strategy labels are split into two groups. On ActorAttack, 9/10 partitions achieve F1 = 1.000; the sole outlier (partition 7, AA F1 = 0.636) uses a 4/5 split that isolates a specific strategy cluster. The near-universal success confirms that domain-relevant binary classification—not specific label semantics—drives cross-attack generalization on template attacks. The AA outlier is consistent with the complexity-dependent boundary observed in Section 4.3: structurally complex attacks are sensitive to specific partition choices while template attacks are not.

M.2 Topic Classification Failure Analysis

Table 18: Per-turn Fisher ratio (mean across all dimensions) and silhouette score between jailbreak and benign turn embeddings (`[CLS]` representations). Higher values indicate greater per-turn separability.

Model	Dataset	Fisher (mean)	Silhouette
Vanilla	IID test	0.118	0.034
Vanilla	DD OOD	0.401	0.411
Vanilla	AA OOD	0.408	0.330
9-class DAPT	IID test	0.073	0.048
9-class DAPT	DD OOD	0.200	0.079
9-class DAPT	AA OOD	0.832	0.301
Topic DAPT	IID test	0.108	0.076
Topic DAPT	DD OOD	0.653	0.274
Topic DAPT	AA OOD	0.311	0.141

Topic DAPT achieves *higher* per-turn Fisher separability than 9-class DAPT on IID and DD (0.653 vs. 0.200 on DD) yet produces far worse conversation-level detection (F1 = 0.541 vs. 0.994). This rules out per-turn separability destruction as the failure mechanism. The 9-class model instead greatly expands the embedding space (centroid L_2 distance 7.5–27.6 vs. 0.9–1.1 for vanilla) while distributing both classes broadly, creating richer representation geometry that the downstream GRU exploits over the turn sequence. Topic classification teaches features that separate turns individually but are temporally inconsistent across a conversation—high per-turn separability does not guarantee that the GRU can learn a stable conversation-level decision boundary.

N Reproducibility Details

Stage-1 training. All variants use DeBERTa-v3-base (184M parameters) with learning rate 2×10^{-5}, AdamW optimizer (β_1=0.9, β_2=0.999, weight decay 0.01), linear warmup over the first 10% of steps, batch size 16, and 4 epochs. MLM variants use 15% random token masking. Training takes approximately 30 minutes per variant on a single NVIDIA A100 (40GB).

Stage-2 training. The BiGRU has hidden dimension 256, 2 layers, and dropout 0.3 (bidirectional, so output dimension 512). The MLP classifier has layers $512 \to 256 \to 2$ with ReLU activation. Training uses learning rate 1×10^{-3}, Adam optimizer, batch size 32, and 20 epochs with early stopping (patience 5) on validation F1. Stage-2 training completes in under 5 minutes per variant.

Data generation. Jailbreak conversations are generated using Qwen3-8B with attack-specific prompt templates. Crescendo conversations follow a 5–8 turn gradual escalation structure. FITD conversations use a 2–3 turn commitment-consistency structure. Deceptive Delight conversations embed harmful content within emotionally engaging 3–5 turn sequences. Benign conversations are generated with topic-anchored prompts across five domains (technology, health, finance, education, general) with matched turn counts.

Strategy annotation. Each user turn is annotated by Qwen3-8B using a structured prompt that provides definitions and two examples for each of the nine strategy categories. The prompt instructs the model to output a single category label. We plan to release all generation and annotation prompts, the full dataset, and training code upon acceptance.

Adversarial paraphrasing. The paraphrased test set is generated by instructing Qwen3-8B to rephrase each user turn while preserving the communicative intent. The resulting paraphrases have mean BERTScore 0.92 and mean lexical overlap 0.20 with the originals. These metrics confirm substantial surface-level variation.

Cross-checkpoint variance. To disentangle Stage-1 from Stage-2 variance, we evaluate three independently trained 9-class DAPT checkpoints (seeds 42, 123, 456) with a fixed downstream BiGRU seed (42). Results: IID F1 = 1.000±0.000, DD OOD = 0.926±0.026, AA OOD = 0.935±0.013. Stage-1 training is stable (OOD std $\leq$ 2.6 pp), confirming that the variance reported in the main paper is not dominated by Stage-1 initialization.

O End-to-End Fine-Tuning Upper Bound

Table 19: Frozen probing vs. end-to-end (E2E) fine-tuning for DeBERTa-v3-base 9-class (3-seed mean±std). E2E unfreezes the Stage-1 encoder during Stage-2 training. Frozen values use main-experiment checkpoints (matching Tables 2–3); Table 25 uses plan_002 checkpoints, yielding higher frozen accuracy (DD = .997, AA = .972) and a narrower E2E gap (≤2.5 pp).

Config	IID	DD OOD	AA OOD
Frozen probing	1.000±.000	.994±.005	.939±.047
E2E fine-tuning	1.000±.000	1.000±.000	**.997±.005**
Δ	.000	+.006	+.058

End-to-end fine-tuning unfreezes the Stage-1 encoder during Stage-2 BiGRU training, allowing gradient flow through the full pipeline. On IID and OOD DD, the two configurations are indistinguishable; on structurally complex ActorAttack, E2E recovers +5.8 pp. The modest gain confirms that frozen probing captures the vast majority of discriminative information encoded by Stage-1 DAPT, and that the remaining OOD gap in the frozen setting reflects downstream optimization constraints rather than missing representational capacity.

P Real-World False Positive Rate

Table 20: False positive rate (%) on 226 WildChat benign multi-turn conversations (GPT-3.5 users, avg 7.4 turns per conversation). All models trained on synthetic Qwen3-8B data (1,000-conversation scale). 3-seed mean±std.

Variant	Seed 42	Seed 123	Seed 456	Mean±Std
9-class	66.8	61.1	76.5	68.1±6.4
MLM	36.7	64.6	32.3	44.5±14.3
Vanilla	47.3	42.5	54.4	48.1±4.9

Table 20 evaluates false positive rates on 226 WildChat benign multi-turn conversations from real GPT-3.5 users (avg 7.4 turns). All variants produce FPR far exceeding deployable thresholds: 9-class is worst at 68.1%, followed by vanilla (48.1%) and MLM (44.5%). Domain adaptation *increases* FPR relative to vanilla (+20 pp for 9-class), indicating that continued pretraining amplifies dependence on the training benign distribution. This is consistent with the cross-LLM false positive spike (Table 12): models trained on synthetic Qwen3-8B benign text implicitly treat unfamiliar conversational styles as jailbreak indicators. High MLM seed variance (std = 14.3%) suggests that unsupervised pretraining is particularly unstable under benign distribution shift. Deployment requires per-domain benign calibration or mixed-source benign training data.

Post-hoc calibration. ROC AUC = 0.700±0.054 indicates insufficient discrimination between WildChat benign and jailbreak conversations at the representation level. At TPR≥95%, the optimal threshold still yields FPR = 62.7%, far above deployable levels. Temperature scaling cannot improve this trade-off, as ROC curves are invariant to monotonic score transforms. The AUC ceiling reflects genuine domain shift: WildChat benign and jailbreak representations overlap substantially, and no post-hoc calibration can overcome this; in-domain benign adaptation is required.

Second benign source: OASST2. To test whether benign-side shift generalizes beyond WildChat, we evaluate on 200 benign multi-turn conversations from OpenAssistant OASST2 (Köpf et al., 2023)—a crowdsourced dataset of real human conversations, distinct from WildChat (GPT-3.5 user logs) in both authorship and topic distribution.

Table 21: OASST2 benign adaptation curve: FPR on 200 OASST2 benign conversations and OOD jailbreak detection F1 as a function of N OASST2 conversations added to training. Seed 42. Compare WildChat (Table 20): FPR reaches 0% at $N = 10$.

N	FPR (%)	IID F1	DD F1	AA F1	FITD F1
0	57.5	–	–	–	–
10	20.5	1.000	1.000	1.000	1.000
25	**1.5**	1.000	1.000	0.953	1.000
50	**0.5**	1.000	1.000	0.990	1.000

Baseline FPR on OASST2 (57.5%) is comparable to WildChat (68.1%), confirming that benign-side shift is not source-specific but a general consequence of synthetic-only benign training. The adaptation curve follows the same pattern—monotonic FPR reduction with preserved OOD detection—but requires more samples: 25 conversations reduce FPR to 1.5% (vs. 0% at $N = 10$ for WildChat), and 50 conversations reach 0.5%. The slower convergence likely reflects greater distributional distance between crowdsourced OASST2 conversations and the synthetic training distribution compared to WildChat's GPT-3.5 user logs.

P.1 Estimated Dual-OOD Detection Performance

The OOD detection F1 reported in the main paper evaluates attack-side OOD with IID benign examples (FPR = 0), providing an upper bound. To estimate performance when *both* attack and benign inputs are OOD, we combine per-seed attack recall (from OOD evaluation) with per-seed benign FPR (from WildChat/OASST2 experiments), assuming independence between attack detection and benign false positives. For each seed, we compute binary detection F1 from the resulting TP, FP, FN counts on the OOD test composition (80 jailbreak + 38 benign).

Table 22: Estimated dual-OOD binary detection F1 under different benign conditions (3-seed mean±std). Attack recall held fixed from OOD evaluation; only benign FPR varies. IID row corresponds to the benign-IID upper bound (FPR = 0).

Benign condition	DD OOD F1	AA OOD F1
IID (FPR = 0)	.994±.009	.981±.019
WildChat, no adapt	.855±.020	.842±.012
OASST2, no adapt	.874±.009	.861±.018
WildChat + 10-shot	.994±.009	.981±.019
OASST2 + 25-shot	.991±.009	.978±.018

Without benign adaptation, dual-OOD F1 drops by 12–14 pp (DD: .994 → .855; AA: .981 → .842), driven entirely by false positives on OOD benign inputs. After few-shot adaptation—10 WildChat or 25 OASST2 conversations—dual-OOD F1 recovers to within 0.3 pp of the IID upper bound, confirming that the lightweight adaptation procedure (Section 4.9) effectively closes the benign distribution gap without compromising attack-side performance.

Q BERT-base Proxy Baseline

Table 23: BERT-base-uncased (110M) frozen probing without DAPT, matching DeepContext's encoder family (Albrethsen et al., 2026). 3-seed mean±std. DeBERTa-v3 results from Table 4 for comparison.

Encoder	IID	DD OOD	AA OOD
BERT-base (110M), no DAPT	.996±.006	.318±.031	.275±.027
DeBERTa-v3 (184M), no DAPT	1.000±.000	.312±.043	.258±.000
DeBERTa-v3 (184M), 9-class	1.000±.000	.994±.005	.939±.047

Table 23 compares BERT-base-uncased (110M parameters, matching DeepContext's encoder) against DeBERTa-v3-base (184M), both without domain-adaptive pretraining. The two vanilla encoders achieve comparable OOD performance (DD: 0.318 vs. 0.312; AA: 0.275 vs. 0.258), despite a 1.7× parameter difference and different pretraining procedures. Both collapse on OOD evaluation while 9-class DAPT recovers to 0.994 (DD) and 0.939 (AA). This confirms that cross-attack generalization is driven by domain adaptation, not encoder architecture or capacity at this scale.

R Confidence Calibration

Table 24: Expected calibration error (ECE, 10 bins) for 9-class and vanilla on IID and OOD evaluation (3-seed mean±std). Lower is better.

Variant	IID	DD OOD	AA OOD
9-class	≈0.000	0.143±0.053	0.116±0.053
Vanilla	0.009±0.007	0.669±0.004	0.504±0.066

The 9-class model achieves substantially lower ECE than vanilla on OOD data (0.12–0.14 vs. 0.50–0.67), indicating that domain adaptation improves not only discriminative performance but also confidence calibration. IID calibration is near-perfect for both variants. The residual OOD ECE (0.12–0.14) suggests that temperature scaling could further improve deployment reliability.

R.1 ToxicChat False Positive Confidence

On ToxicChat, 9-class false positives are overwhelmingly high-confidence: 70.2% of majority-vote FPs exceed $p > 0.9$ (median 0.999), compared to 58.8% for JB-MLM (median 0.933). Per-seed analysis is more extreme—94–98% of single-model 9-class FPs are high-confidence (median 1.0), with the borderline fraction (25%) in majority vote arising from cross-seed disagreement rather than individual model uncertainty. This confirms that the 9-class ToxicChat FPR reflects systematic feature misalignment—classification DAPT shifts benign text into the jailbreak region with high confidence—rather than boundary ambiguity.

S Frozen Probing Verification

Table 25 compares frozen probing across DAPT objectives and against end-to-end fine-tuning. On easy OOD (DD, FITD), frozen MLM and frozen 9-class representations perform comparably ($\Delta \leq 1.3$ pp). On hard OOD (ActorAttack), 9-class retains a modest advantage (0.972 vs. 0.941, $\Delta = 3.1$ pp) with lower variance ($\sigma = 0.026$ vs.

Table 25: Frozen probing vs. end-to-end (E2E) fine-tuning (3-seed mean±std). 9-class: plan_002 checkpoints; MLM: plan_017_mlm_control (matched training). Frozen probing trains only BiGRU+MLP on frozen DAPT embeddings.

Config	IID	DD OOD	AA OOD	FITD OOD
Frozen 9-class	1.000±.000	.997±.005	.972±.026	1.000±.000
Frozen MLM	.982±.016	.993±.009	.941±.037	.987±.009
E2E 9-class	1.000±.000	1.000±.000	.997±.005	

Table 26: Full-conversation vs. last-turn-only F1 (3-seed mean). Truncation widens the 9-class advantage, ruling out multi-turn structure as the driver of the ToxicChat reversal.

Dataset	Mode	9-class	JB-MLM
IID	Full	1.000	.983
IID	Last turn	.991	.894
DD OOD	Full	.997	.993
DD OOD	Last turn	.942	.924
AA OOD	Full	.972	.941
AA OOD	Last turn	.978	.927
FITD OOD	Full	1.000	.987
FITD OOD	Last turn	1.000	.927

Mean Δ (9-class − MLM): Full +1.6 pp, Last turn +6.0 pp. 9-class advantage widens under truncation across all families, indicating the reversal on ToxicChat is driven by distributional differences between synthetic and human text, not by multi-turn structural dependence.

Table 27: Representation geometry and class separability across Stage-1 variants. PR: participation ratio (effective dimensionality of the embedding space). Fisher ratio: inter-class distance^2 / sum of intra-class variances. Weight Δ: L2 distance from pretrained weights (relative to pretrained norm).

Variant	PR	Fisher	Wt. Δ (%)
Vanilla	31.9	<0.001	0.000
Wiki-MLM	25.0	<0.001	0.221
JB-MLM	24.6	<0.001	0.256
Topic	14.5	0.012	0.371
Scrambled	7.8	2.60	0.283
9-class	9.4	4.47	0.303
Binary	3.7	19.7	0.239

0.037). Both far exceed vanilla encoders (≤0.318; Table 23). The E2E advantage over frozen 9-class is ≤2.5 pp, confirming that Stage-1 representation quality is the primary source of OOD generalization.

A striking asymmetry emerges when comparing frozen probing to the full pipeline (Table 3): MLM frozen representations *outperform* the full MLM pipeline on AA OOD (0.941 vs. 0.893), while 9-class frozen exactly matches its full pipeline (0.972 vs. 0.972). This indicates that MLM's cross-seed instability originates in Stage-2 BiGRU training rather than Stage-1 DAPT: the MLM encoder produces adequate representations that are then degraded by unstable downstream optimization. Classification DAPT, by contrast, produces representations that remain robust through Stage-2 training, explaining its consistently lower variance (Table 3).

This two-level structure (DAPT vs. no-DAPT producing a representation quality leap of +70 pp; classification vs. MLM DAPT producing a stability difference) is consistent with two further observations: λ=0 matches $\lambda > 0$ during E2E fine-tuning (Table 35), and mean pooling matches BiGRU under E2E (Table 30). Once DAPT representations are strong, downstream configuration details have minimal impact.

S.1 Single-Turn Truncation Analysis

To isolate whether the ToxicChat evidence-hierarchy reversal (Section 5) reflects multi-turn structural dependence, we truncate each synthetic test conversation to its last user turn and re-evaluate both 9-class and JB-MLM models (3-seed mean).

Representation geometry analysis. To characterize the mechanism behind classification DAPT's robustness, we analyze Stage-1 representations across all seven DeBERTa-v3-base variants (seed 42). We extract `[CLS]` embeddings from 150 test conversations (50 IID + 50 DD OOD + 50 AA OOD) and compute three metrics.

Classification variants compress representations into 2–9 effective dimensions (vs. 25–32 for MLM/vanilla) while increasing Fisher ratio by 3–5 orders of magnitude. Weight distances from pretrained are comparable across all variants (∼0.2–0.4%), indicating that the mechanism is representational restructuring rather than parameter-space regularization. Scrambled labels produce similar compression (PR = 7.8, Fisher = 2.6), confirming that the classification objective structure—not label semantics—drives discriminative compression.

Layer-wise localization. To determine *where* discriminative compression occurs, we compute PR and Fisher ratio at every DeBERTa-v3-base layer (0–12) for four variants (9-class, scrambled, JB-MLM, vanilla) using the same 236-sample evalua-

tion set (seed 42).

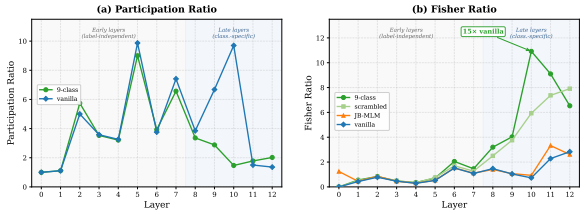


Figure 7: Layer-wise representation analysis. (a) Participation ratio: classification variants (9-class, scrambled) show sharp dimensional collapse at layers 9–10 (PR $\approx$ 1.5–2.9) while vanilla/JB-MLM maintain high dimensionality (PR $\approx$ 7–10). (b) Fisher ratio (log scale): class separability diverges from layer 8, with 9-class peaking at 10.9 (layer 10) vs. vanilla 0.73 (15×).

Layers 0–7 show no systematic divergence across variants (Fisher < 2.1 for all; PR tracks similar oscillation patterns), consistent with Maennel et al. (2020)'s finding that early layers learn label-independent features. From layer 8 onward, classification-supervised variants diverge sharply: 9-class Fisher rises from 3.2 (layer 8) to 10.9 (layer 10, peak), while vanilla remains below 1.5 throughout. Simultaneously, 9-class PR drops from 3.4 to 1.5 at layer 10—a 6.6× compression relative to vanilla (PR = 9.7). JB-MLM tracks vanilla across all layers (Fisher ≤ 3.3, PR ≥ 1.4), confirming that MLM does not induce discriminative restructuring. Scrambled labels exhibit the same late-layer pattern (Fisher peaks at 7.9, layer 12; PR drops to 1.5, layer 12), reinforcing that any classification signal—regardless of label semantics—triggers late-layer compression.

Discriminative subspace probe. To test whether the compressed subspace is *sufficient* for OOD detection, we project layer-10 `[CLS]` representations onto the top-k SVD components and train a logistic regression probe (3 seeds, sklearn default). Controls: random k-dimensional projections (10 random bases, mean reported) and vanilla/MLM models under identical projection.

Three findings establish a link between compression and generalization (Table 28). First, 9-class top-3 achieves perfect ActorAttack detection (F1 = 1.000) using only 3 of 768 dimensions—all discriminative information resides in the compressed subspace. Second, random 3-dimensional projections degrade to 0.928 on AA, confirming that the specific SVD directions matter. Third, JB-MLM reaches only 0.636 on AA even with all 768 dimensions, demonstrating that without

Table 28: Discriminative subspace probe: OOD F1 when projecting layer-10 representations onto top-k SVD components vs. random k-dimensional projections. 9-class top-3 achieves perfect AA detection with 3/768 dimensions; MLM fails even with all 768.

Model	**Projection**	DD	AA	FITD
9-class	top-3	1.000	**1.000**	1.000
9-class	top-5	1.000	1.000	1.000
9-class	random-3	1.000	0.928±.129	1.000
9-class	random-5	1.000	0.952	1.000
9-class	full-768	1.000	1.000	1.000
scrambled	top-3	1.000	0.962	1.000
scrambled	random-3	0.981	0.766	0.987
JB-MLM	top-3	1.000	0.618	1.000
JB-MLM	full-768	1.000	0.636	1.000
JB-MLM	random-5	0.685	0.584	0.744

classification-induced compression, the representation lacks the structure needed for cross-attack generalization. DD and FITD are near-ceiling for all models, making ActorAttack the critical test. SVD directions computed on IID training data only (without OOD test exposure) yield equal or higher F1 than all-data SVD (AA: 0.947 vs. 0.904), confirming the subspace is not an artifact of test-set leakage.

SVD necessity ablation. The probe above establishes *sufficiency*: 3 SVD dimensions carry enough information. To test *necessity*, we project *out* the top-3 SVD directions from full 768-dimensional 9-class `[CLS]` embeddings and re-run BiGRU classification on OOD data (3 seeds). A random-3 removal control verifies that the effect is specific to the discriminative subspace. The top-3 directions explain 89.8% of training-set variance.

Table 29: SVD necessity ablation: OOD F1 after projecting out top-3 vs. random-3 SVD directions from 9-class layer-10 embeddings (3-seed mean±std).

Condition	DD OOD	FITD OOD	AA OOD
Original 768-dim	1.000±.000	1.000±.000	.994±.005
Remove top-3 SVD	.831±.064	1.000±.000	.938±.011
Remove random-3	1.000±.000	1.000±.000	.994±.005

Removing top-3 SVD directions degrades DD detection by 17 pp (F1: 1.0 $\rightarrow$ 0.831) and AA by 5.6 pp (0.994 $\rightarrow$ 0.938), while random-3 removal has zero effect. FITD is unaffected by either ablation, indicating its discriminative information is more distributed. Combined with the sufficiency result (Table 28), the top-3 SVD directions are both sufficient and necessary for DD detection, suffi-

cient but partially redundant for AA, and sufficient but not necessary for FITD.

T Temporal Aggregation Comparison

Table 30: Temporal aggregation comparison: mean pooling vs. BiGRU under frozen and end-to-end (E2E) settings (3-seed mean±std). Plan_002 9-class checkpoints.

Setting	Aggregator	DD OOD	AA OOD
Frozen	BiGRU	.997±.005	.972±.026
	Mean pooling	.991±.013	.962±.015
E2E	BiGRU	1.000±.000	.997±.005
	Mean pooling	1.000±.000	1.000±.000

Table 30 compares BiGRU and mean pooling under both frozen and E2E settings. Under frozen probing (the main paper's pipeline), BiGRU provides a marginal advantage on ActorAttack (+1.0 pp) with overlapping confidence intervals, consistent with the main text finding that temporal order is not a core detection factor ($|\Delta| < 1$ pp under turn shuffling; Section 4.3). Under E2E fine-tuning, the difference vanishes entirely: mean pooling saturates all benchmarks at F1 = 1.000. The choice of frozen BiGRU in the main pipeline reflects a design trade-off: it enables representation-level analysis (CKA, t-SNE) that E2E would confound, while sacrificing $\leq$1 pp on the hardest OOD set.

U Length Stratification Analysis

Table 31: Jailbreak recall by conversation length bin on OOD evaluation (3-seed majority vote). Benign conversations span 12–20 turns; jailbreak conversations span 3–10 turns.

Dataset	Variant	1–3 turns	4–6 turns	7–10 turns
DD OOD	9-class	—	100.0%	99.4%
	Vanilla	—	$\leq$2.6%	$\leq$1.7%
AA OOD	9-class	93.3%	95.6%	100.0%
	Vanilla	0.0%	$\leq$2.6%	0.0%

Table 31 stratifies jailbreak recall by conversation length. The 9-class model shows no systematic degradation: ActorAttack recall increases monotonically from 93.3% (1–3 turns) to 100% (7–10 turns), and DD recall is near-perfect across all bins.

Length confound. Benign conversations (12–20 turns) and jailbreak conversations (3–10 turns) have zero length overlap in the OOD evaluation sets, raising the concern that the model discriminates by conversation length rather than content. Three observations argue against this interpretation: (1) Vanilla DeBERTa has access to identical length information via the BiGRU but achieves recall $\leq$2.6% across all bins; length alone is insufficient without domain-adapted representations. (2) At prefix K=1, where all conversations are length-equalized, 9-class still achieves F1 = 0.773±0.041 on ActorAttack (Table 8), demonstrating content-based detection independent of length. (3) The model is trained on IID data (Crescendo) and cannot learn OOD-specific length distributions; generalization to unseen length patterns must rely on content features that transfer across attack families.

Table 32: Turn-count independence verification (9-class DeBERTa-v3-base, same checkpoint as Table 2, 3 seeds). Evaluated on the full OOD test set (80 jailbreak + 75 benign conversations) rather than the balanced subset used in Table 2. Part A: FPR on test benign truncated to K user turns (all seeds FPR = 0). Part B: WildChat FPR by turn-count bucket (mean±std). Part C: jailbreak detection (F1) after padding with benign turns to 15+ total turns (mean±std where available).

Condition	FPR / F1	n
Part A: Benign truncation (test set)		
Original (12–20 turns)	FPR = .000	38
Truncated K=3	FPR = .000	38
Truncated K=5	FPR = .000	38
Truncated K=8	FPR = .000	38
Truncated K=10*	FPR = .000	38
Part B: WildChat FPR by turn count		
Short (3–10 turns)	FPR = .770±.105	196
Medium (11–15 turns)	FPR = .942±.054	23
Long (16+ turns)	FPR = .857±.117	7
Part C: Jailbreak padding to 15+ turns		
DD original	F1 = .996	80
DD padded	F1 = .971 (Δ=−2.5 pp)	80
AA original	F1 = .963	80
AA padded	F1 = .383±.261 (Δ=−57.9 pp)	80

* K=10 did not truncate any conversation (test benign max ∼8 user turns).

Direct empirical verification. Truncating benign conversations to match jailbreak turn counts (Part A) produces zero false positives across all truncation levels and all 3 seeds, directly confirming that the model does not use conversation length as a detection shortcut. K=3 and K=5 are the effective control conditions where all 38 conversations were truncated; K=10 exceeds the maximum test benign length and serves as a no-op control.

WildChat FPR by turn count (Part B) provides

complementary evidence: short conversations (3–10 turns, n=196) have *lower* FPR than medium-length ones (11–15 turns, n=23), inconsistent with a "short = jailbreak" shortcut. Medium and long buckets have limited sample sizes and should be interpreted as directional evidence only.

Padding jailbreak conversations with benign turns (Part C) reveals an asymmetry: DD detection is resilient (Δ=−2.5 pp), but ActorAttack collapses (Δ=−57.9 pp). Padding experiments append benign turns *after* the complete attack sequence; in practice, adversaries would not dilute their own successful attacks. The collapse nonetheless exposes BiGRU temporal aggregation as vulnerable to attention dilution on subtle attack signals, a limitation discussed in Section 7.

V Inference Efficiency

Table 33: Inference efficiency comparison. DeBERTa+BiGRU and TF-IDF+LR measured on IID test (76 conversations, avg 8.3 turns, single A100). LLM latency from separate OOD evaluation (118 conversations; exp10); cross-condition comparison is indicative only.

Method	Latency (ms)	Params	VRAM
TF-IDF+LR	0.43±0.05	∼10K	CPU
DeBERTa+BiGRU (ours)	12.1±1.2	186.7M	862 MB
Qwen3-8B 3-shot†	∼5,200	8B	∼16 GB

†Qwen3-8B latency from exp10 (DD/AA OOD, 118 conversations); not directly comparable to encoder latency measured on IID test.

Table 33 compares inference cost. On matched data (IID test), DeBERTa+BiGRU processes a conversation in 12.1 ms, 28× slower than TF-IDF (0.43 ms), but this speed gap purchases +71 pp on ActorAttack (F1 = 0.972 vs. 0.261; Table 3). LLM-as-judge approaches are orders of magnitude slower (∼5.2 s per conversation for Qwen3-8B 3-shot) while achieving substantially lower OOD detection (DD = 0.566, AA = 0.652; Table 6). Zero-shot Qwen3-8B achieves F1 = 0.0 on DD OOD because it predicts all conversations as benign (Table 6). The full DeBERTa+BiGRU pipeline requires 862 MB GPU memory, enabling deployment on consumer hardware. The 24.3× speedup reported in the main text derives from a matched-condition comparison (exp46: DisPeD 13.12 ms vs. Qwen3-8B 318.53 ms on the same GPU and data); the larger gap in Table 33 reflects different hardware conditions and evaluation sets.

W Data Efficiency

Table 34: Data efficiency of 9-class DAPT (3-seed mean±std, sample std with $n-1$ divisor) and vanilla baseline (seed 42 only). N = total training conversations (balanced jailbreak/benign).

N	**Variant**	IID	DD OOD	AA OOD
50	9-class	1.000±.000	1.000±.000	.969±.037
125	9-class	1.000±.000	1.000±.000	.975±.019
250	9-class	1.000±.000	1.000±.000	.975±.035
350	9-class	1.000±.000	.978±.030	.972±.033
50	Vanilla	.646	.529	.234
125	Vanilla	.532	.244	.244
250	Vanilla	.987	.365	.502
350	Vanilla	1.000	.360	.489

Table 34 shows that 9-class DAPT reaches near-saturation performance with as few as 50 conversations (25 jailbreak + 25 benign): AA OOD F1 = 0.969±0.037, already surpassing vanilla at full training size (N=350, AA = 0.489) by 48 pp using only 14% of the data. DD OOD is perfectly solved at all training sizes. Vanilla performance is noisy across N (single seed); the consistent pattern is that vanilla never exceeds DD = 0.53 or AA = 0.50 regardless of training size, while 9-class maintains AA $\geq$ 0.969 at all N. Seed 456 is systematically lower across all N values, contributing most of the cross-seed variance; this is consistent with the initialization sensitivity observed in the main results.

X Auxiliary Loss Weight Ablation

Table 35: Effect of auxiliary persuasion loss weight λ during Stage-2 E2E fine-tuning (seed 42). Stage-1 DAPT encoder is identical across all rows; only the Stage-2 loss $L = L_{\text{binary}} + \lambda \cdot L_{\text{strategy}}$ varies.

λ	IID	DD OOD	FITD OOD	AA OOD
0	1.000	1.000	1.000	1.000
0.1	1.000	1.000	1.000	.990
0.5	1.000	1.000	1.000	.981
1.0	1.000	1.000	1.000	.981
2.0	1.000	1.000	1.000	.944

During Stage-2 E2E fine-tuning, the auxiliary persuasion loss weight λ has limited impact on detection (Table 35): the maximum AA OOD gap across all λ values is 5.6 pp (single seed), comparable to the cross-seed variance (∼6 pp) observed in the main results. All configurations converge to perfect validation F1 within epoch 1, indicating that DAPT representations already encode sufficient

discriminative features for downstream classification (Section S). The value of persuasion strategy classification lies in Stage-1 DAPT, where it shapes encoder representations that transfer across attack families, not in the Stage-2 auxiliary loss.

Y OOD Strategy Prediction Analysis

Table 36: OOD strategy prediction accuracy (jailbreak turns only) and detection rate for the 9-class model (3-seed mean). r: Pearson correlation between per-strategy prediction accuracy and per-strategy detection rate.

Dataset	Strat. Acc.	Det. Rate	Missed (per seed)	r
DD OOD	50.1%	99.6%	0 / 0 / 1	∼0.45
AA OOD	50.1%	96.2%	1 / 7 / 1	∼0.24

Table 36 reports OOD strategy prediction accuracy alongside detection rates. Strategy prediction accuracy is substantially above random (50.1% vs. 11.1% chance for 9 classes) and positively correlated with detection success ($r \approx 0.24$–0.45), but strategy prediction is not the detection mechanism: frozen probing with a randomly initialized strategy head (Appendix S) achieves 97.2% AA OOD detection, and λ=0 (no strategy loss) matches or exceeds $\lambda > 0$ during E2E fine-tuning (Table 35). The classification objective shapes representations during Stage-1 DAPT training; at inference, strategy predictions are a byproduct of those representations, not a driver of detection.

Z Cross-Seed Ensemble

Table 37: Majority vote ensemble (3 seeds) vs. single-seed mean on OOD Full. "Fixed" = samples corrected by ensemble; "Broke" = samples corrupted by ensemble.

Variant	DD single→ens.	AA single→ens.	Fixed	Broke
9-class	.997→**1.000**	.972→**.981**	5	0
MLM	.914→**1.000**	.897→**.953**	26	0
Vanilla	.258→.258	.258→.258	0	0

Table 37 shows that majority vote never corrupts correct predictions (Broke = 0 for all variants). MLM benefits most from ensembling (+8.6 pp DD, +6.0 pp AA), because its per-seed errors are complementary: each seed fails on different samples, and the majority corrects these. This further exposes MLM's seed instability: its individual seeds are unreliable, but their errors are uncorrelated. By contrast, 9-class seeds are already highly consistent (std $\leq$0.026), leaving minimal room for ensemble improvement (+0.3–0.9 pp). Vanilla errors are systematic (all seeds fail on the same OOD samples), so ensembling provides no benefit.

Z.1 Seed-Level F1 Comparisons

To verify that cross-seed pooling does not mask inconsistent pairwise orderings, Table 38 reports per-seed F1 for all variants on OOD DD and ActorAttack at full conversation length. The overall evidence hierarchy is maintained across seeds, with 9-class and binary consistently outperforming MLM; scrambled shows one reversal on AA (seed 42).

Table 38: Per-seed F1 on OOD Full for key variants (seeds 42, 123, 456). The relative ordering is consistent across seeds.

	DD-Full			AA-Full		
Variant	s42	s123	s456	s42	s123	s456
9-class	1.000	1.000	.990	.990	.935	.990
Binary	1.000	1.000	1.000	.981	.990	1.000
Scrambled	.990	1.000	1.000	.891	.990	.971
MLM-only	1.000	.990	.751	.971	.925	.784
Vanilla	.285	.305	.348	.258	.258	.258